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P CYGNI PROFILES IN LONG-PERIOD VARIABLES

FRED SANNER
Kitt Peak National Observatory*
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ABSTRACT

P Cygni profiles have been observed in the Na I and K I resonance lines of χ Cygni at maximum and postmaximum phases. These features indicate the presence of a region in the Mira atmosphere which can be termed circumstellar in the same sense as used to describe the envelopes surrounding earlier M type giants and supergiants. The potential use of these profiles in setting limits on the stellar heliocentric velocity and on the expansion velocity of the circumstellar gas is discussed.

Subject headings: stars: mass loss — stars: long-period variables — stars: circumstellar shells

I. INTRODUCTION

P Cygni profiles are commonly found in the cores of strong optical resonance lines in normal M giants and supergiants. It is believed that these features are formed in an expanding circumstellar gas envelope which scatters the photospheric radiation incident upon it. This circumstellar region appears to have a number of distinct geometric and kinematic properties which distinguish it from the region in which the stellar absorption line spectrum is formed (Deutsch 1960; Sanner 1976).

It is not clear, however, what properties describe the circumstellar region surrounding the Mira variables. Circumstellar molecular and dust emission has been detected in these stars (Schwartz and Barrett 1970; Wilson, Barrett, and Moran 1970; Kaifu, Buhl, and Snyder 1975; Gillett, Merrill, and Stein 1971; Gehrz and Woolf 1971), features which are weak or absent in the spectra of normal M giants. Their presence indicates that the physical conditions in Mira envelopes differ somewhat from those in the envelopes of earlier M stars. Their strengths require larger envelope column densities, so that strong circumstellar features are expected in the optical resonance lines.

The optical spectra of Mira variables are so complex, however, that it is not always possible to identify circumstellar features unambiguously. Severe blending, and in some stars line splitting, contribute to the identification problem. Also, systems of shortward-displaced emission lines, which first appear in these stars near maximum light, often coincide with the expected positions of the circumstellar features. The problem of identification may, however, be more than one of simple detection. The line splitting and emission features may be intimately related to the circumstellar phenomenon; perhaps they are formed in a circumstellar region themselves.

The purpose of this *Letter* is to report observations of P Cygni features in the Na I and K I resonance lines of χ Cyg at maximum and postmaximum phases. These features indicate the presence of an outer, extended region in the Mira atmosphere which can be termed circumstellar in the same sense as used to describe the envelopes of earlier M stars. In particular, the profiles indicate that the envelope extends at least several stellar radii above the region in which the local continuum is formed and scatters the resonance line radiation incident upon it (in this way giving rise to the observed P Cygni emission). It is possible to estimate the heliocentric velocity of the central star and the expansion velocity of the circumstellar gas from measurements of the profiles, provided certain physical conditions prevail in the envelopes.

II. OBSERVATIONS

Chi Cygni was observed on 1976 July 10, seven days after the predicted light maximum (Mattei 1976), and again on 1976 September 12, with the Cassegrain echelle spectrograph on the 4 m Mayall telescope at Kitt Peak National Observatory. Spectra were obtained on baked IIIa-J plates without filter in the region of the Na I D lines, and with an RG 610 filter in the region of the K I resonance lines at $\lambda\lambda$7665 and 7699. The dispersion is 2.5 Å mm^{-1} at 5000 Å.

Similar plates of the K I lines were obtained of the comparison stars α Ori and R Leo on these and other nights. The K I λ7699 line of α Ori has been well observed, both at high spectral (Goldberg *et al.* 1975; Sanner 1976) and high spatial (Bernat and Lambert 1975) resolutions. The core of λ7699 in R Leo is observed to be split into two components of approximately equal strength, one at the normal photospheric absorption line position, the other displaced blueward. As line splitting on the same plate is confined to this single line, and as R Leo has no history of line splitting in the literature, this appearance is tentatively interpreted as a P Cygni feature in the line core. The local maximum in intensity separating the two components is interpreted as circumstellar emission. This emission does not rise above the continuum level.

Figure 1 shows the profiles of the Na I D lines of χ Cyg on July 10. These lines were traced with the

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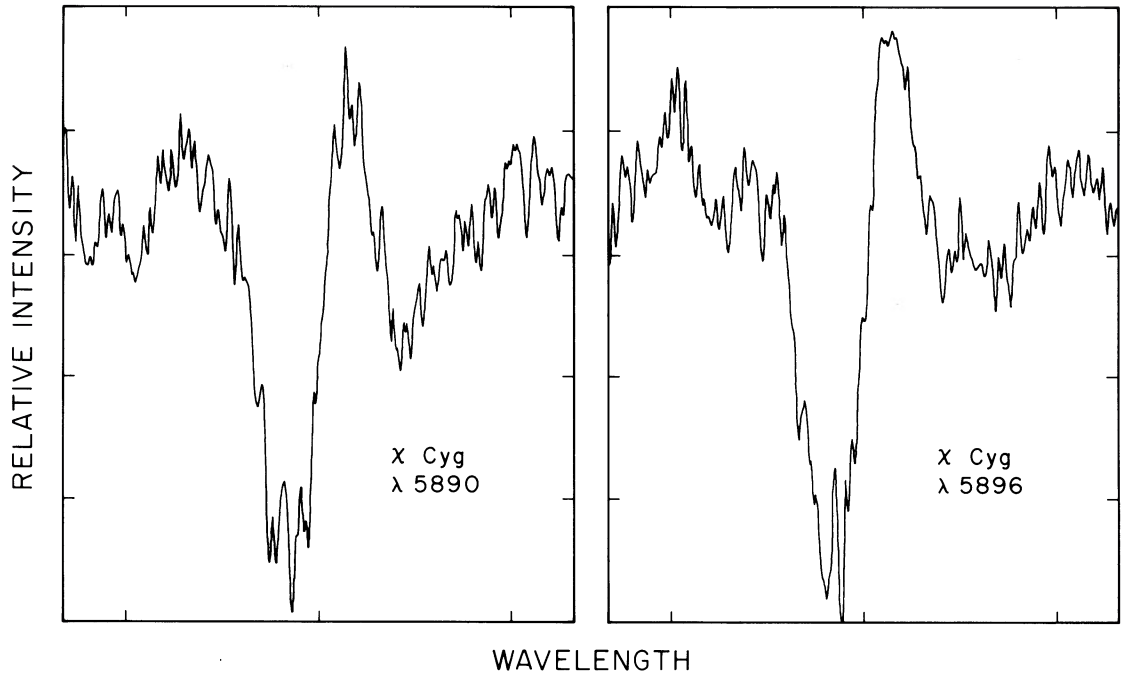

FIG. 1.—Circumstellar P Cygni profiles in the Na I D lines of χ Cyg. The abscissa defines the zero level of intensity. The unit of wavelength is 1 Å.

KPNO PDS microdensitometer in density units, and converted to relative intensity using a spot sensitometer plate developed with the stellar spectrum. Both profiles are presented on a common intensity scale. The red wing of the photospheric line and the P Cygni feature are readily seen in each profile. A circumstellar interpretation of these features is very natural, since Na I and K I are the expected dominant circumstellar lines in this spectral region. Detailed measurements of the K I lines (see below) seem to rule out all other possibilities. The P Cygni character of λ5896 is more pronounced than that of λ5890. Such an apparent inversion in line strengths is possible for circumstellar features formed by scattering, since the amount of radiation scattered by the envelope is determined in part by the intensity in the underlying stellar line profile, which appears to be deeper in λ5890.

The heliocentric velocities of the absorption and emission components of K I λ7699, denoted by $V_{\rm abs}^{\rm cs}$ and $V_{\rm em}^{\rm cs}$, respectively, along with representative velocities of the stellar optical absorption and emission lines, $V_{\rm abs}^{\rm opt}$ and $V_{\rm em}^{\rm opt}$, are given in Table 1. The circumstellar velocities have been measured in each star with a standard error of less than 2 km s^{-1} by fitting polynomials to Grant machine measurements of neighboring atmospheric A band lines. Inspection of this table shows that the P Cygni profile in α Ori is superposed on the stellar absorption line (Goldberg *et al.* 1975; Sanner 1976). In the Mira variables, the P Cygni features lie between the stellar absorption and emission lines. In χ Cyg, the circumstellar absorption component has a heliocentric velocity typical of interstellar lines in this direction of the sky (Adams 1949; Hobbs 1969). Although a significant interstellar contribution may occur in the D lines, the great strengths of the K I absorption components indicate that these lines are primarily circumstellar in origin (Hobbs 1974). The circumstellar emission components of χ Cyg and R Leo are shifted longward of the stellar emission lines, so a common origin with these lines is dismissed. The emission is also shifted relative to the laboratory rest wavelength, so night sky contamination cannot be a dominant contributor. As the emission in χ Cyg rises distinctly above the continuum in all four lines, it is not likely to be an artifact of blending. The profiles are probably also largely unaffected by line splitting (Maehara 1968), for which there is no evidence on the present plates.

III. THE STELLAR HELIOCENTRIC VELOCITY

In addition to revealing various properties of the envelope, the P Cygni profiles are potentially useful in determining the heliocentric velocity of the central star. It has not been possible to determine this velocity from optical observations, in spite of its central importance in interpretations of the atmospheric motions in Mira variables. Reid and Dickinson (1976) have proposed a method for determining this velocity, based on measurements of circumstellar radio emission line profiles. The similar use of the optical P Cygni profiles described below is identical in principle to their method, but is beset by a number of additional complications.

In χ Cyg, the marked circumstellar emission components suggest that the star does not occult a significant portion of the receding half of the envelope. The P Cygni profile is thus formed both in the approaching gas in front of the star and in the receding gas behind

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it, and therefore embraces the stellar center of mass velocity. The exact location of this velocity within the profile is determined by unknown details of the envelope geometry and kinematics. Computed line profiles, which have been used to analyze the circumstellar features in earlier M stars (Sanner 1976), consistently place the center-of-mass velocity between the absorption and emission components, generally nearer the latter. Although it may be possible to find more general geometries and kinematics for which this is not the case, the assumptions underlying these models will be adopted here. The velocity of the point midway between the absorption and emission components therefore serves as an estimate of the stellar center of mass velocity V_*. Similarly, the expansion velocity $V_{\rm exp}$ is estimated to be half the velocity separation between these components.

The models described above do not account for scattering by circumstellar dust, which is observed at 11 μm in the stars in Table 1. Jones (1973) has discussed the apparent displacement of the stellar center-of-mass velocity by scattering from an expanding, pure dust envelope. The more general problem of interest here, the transfer of resonance line radiation through an envelope coextensive in both gas and dust, has not yet been treated. In such an envelope, the optical depths in the centers of strong resonance lines are likely to be much greater than the scattering optical depth of the dust, if reasonable envelope and grain properties are assumed. It is therefore plausible that gas scattering will dominate the frequency diffusion of photons through the envelope at line frequencies. The P Cygni profiles may then be expected to reflect the real gas motions. It is difficult, however, to foresee all of the radiative transfer effects involved, and detailed calculations are very desirable.

The P Cygni center-of-mass and expansion velocities, inferred from the observed profiles with the aid of the gas shell models, are given in Table 1. The center-of-mass velocity of α Ori coincides with the photospheric absorption line velocity within the uncertainties of measurement. Thus, there appears to be no net expansion or contraction of the photosphere of this star in excess of several km s^{-1}. The center-of-mass velocity of R Leo lies midway between the optical absorption and emission lines and coincides with the velocity of the SiO ground-state emission line measured by Reid and Dickinson (1976). The width of the P Cygni profile, which measures the spread in velocities within the envelope, also agrees with the scale of the SiO line. This agreement between optical and radio data reinforces the circumstellar interpretation of each, and appears to rule out any displacement in the position of the optical P Cygni feature resulting from dust scattering. The expected magnitude of this displacement is on the order of $V_{\rm exp}$ (Jones 1973), however, which for R Leo is not substantially larger than the measurement errors. Any definitive conclusions concerning the importance of dust scattering must therefore await further observation. The center-of-mass velocity of χ Cyg also lies between the optical absorption and emission lines. The widths of the P Cygni profiles in this star are noticeably larger than those in R Leo. Ground-state SiO emission has not yet been observed.

It would be of great interest to estimate the mass loss rate from the circumstellar profiles of χ Cyg. Unfortunately, both Na I and K I probably represent trace ionization states of the circumstellar gas. A very uncertain ionization calculation is therefore required to obtain the column densities and mass loss rate. Hopefully, future observations of other lines will lead to quantitative estimates of the mass loss rate and physical conditions in the envelope.

IV. CONCLUSION

P Cygni profiles have been observed in the Na I and K I resonance lines of χ Cyg. These features indicate the presence of a circumstellar gas envelope which extends at least several stellar radii above the level of continuum formation and scatters the resonance line radiation incident upon it. The P Cygni profiles lie between the stellar optical absorption and emission lines. If scattering by circumstellar dust does not affect the formation of the P Cygni features, then the heliocentric velocity of the star can be estimated to lie approximately midway between the circumstellar absorption and emission components.

TABLE 1

VELOCITY DATA*

Star	Spectral Type†	Date (1976)	$V_{\rm abs}^{\rm cs}$	$V_{\rm em}^{\rm cs}$‡	V_*	$V_{\rm exp}$	$V_{\rm abs}^{\rm opt}$§	$V_{\rm em}^{\rm opt}$§
χ Cyg.....	S7e	July 10	−17.7	− 6.5	−12.1	5.6	0	−16
		Sep. 12	−16.6	− 6.1	−11.4	5.3	...	...
R Leo.....	M7e	Apr. 9	+ 4.0	+ 9.4	+ 6.7	2.7	+15	0
α Ori.....	M2 Iab	Apr. 9	+11.6	...	...	...	+22	...
		Sep. 12	+11.9	+26.9	+19.4	7.5	+19	...

* Relative to the Sun, in units of km s^{-1}.

† At maximum light, from Keenan, Garrison, and Deutsch 1974.

‡ Emission component in α Ori absent on 1976 April 9.

§ Data for χ Cyg and R Leo from Wallerstein 1975; α Ori velocities are based on measurements of Mg I $\lambda\lambda$7657.60, 7691.57.

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FRED SANNER: Kitt Peak National Observatory, Box 26732, Tucson, AZ 85726

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